

## **Important Questions**

**Subject:**Database Management Systems

**Program:**BE IT

**Semester:**IV

- 1) Explain data abstraction in brief. Differentiate between physical and logical data independence.
- 2) Explain DDL and DML statements in SQL with suitable examples.
- 3) Define relational data model. Draw an ER-diagram for a Library Management System including primary key, weak entity, composite attribute, derived attribute and multi-valued attributes in your ER-diagram.
- 4) Construct an ER diagram for a Metropolitan Bus Park. There are many gates for entering bus park. Different gates are assigned to different bus routes. A route uses different buses. Buses consist of different seats which are assigned to different passengers. Frequent travelers are also in passenger. Associate a log of reservation date while reserving seats. The passenger's name must have two attributes first\_name and last\_name. Each of entities must have primary key attribute as far as possible. The cardinality mapping should be explained properly.

5) Write SQL statements for the following:

Sailors(sid,sname,rating,age)

Boats(bid,bname,color)

Reserves(sid,bid,day)

Write a SQL statements for the following.

- i) Find the records sailors who have reserved boat number 103(bid=13)
- ii) Update the color of the boat, where bid is 104, into green
- iii) Find the name of sailor who have reserved a red or green boat.
- iv) Find the name of sailors who have reserved boat number 103 on day 5.

6) Consider a simple relational database of Hospital management system.(Underlined attributes represent primary key attributes)

Doctors (Doctor\_ID, DoctorName, Department, Address, Salary)

Patients(PatientID, Patient\_Name, Address, Age, Gender)

Hospitals (PatientID, DoctorID, HospitalName, Location)

Write down the SQL statements for the following:

- i Display ID of patient admitted in hospital at pokhara and whose name ends with 'a'
- ii Delete the records of Doctors whose salary is greater than average salary of doctors
- iii Increase salary of doctors by 18.5% who works in OPD department
- iv Find the average salary of Doctors for each address who have average salary more than 55k.

7) Differentiate between join and sub query. Explain different SQL joins with examples.

8) How does "GROUP BY" clause work? What is the difference between WHERE and HAVING clause? Explain each with examples.

9) Define Stored Procedure. List the advantages and disadvantages of Stored Procedure. Explain how stored Procedure is created with example.

10) Explain Primary Key and Foreign Key in brief. List out the advantages of DBMS over traditional file processing System.